Varunprasaath R. Selvaraju, Ph.D.

(+1) 678-207-9914 | varunprasaath@gmail.com | linkedin.com/in/varunprasaath-rs |

PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in DOE-guided process development, hypothesis-driven experimental design, analytical characterization, and scale-up-oriented technical reporting. Experienced developing process windows, analyzing out-of-trend data, and translating multi-technique results into practical next-step recommendations for cross-functional development teams.

CORE COMPETENCIES

Process Development | DOE & Experimental Design | Data Analysis | Technical Reporting | Scale-Up Support

Analytical Characterization | SOP-Ready Documentation | Cross-Functional Communication

WORK EXPERIENCE

University of Colorado Boulder — July 2025 - Present

Postdoctoral Researcher

- Lead chemical process development and optimization for bio-based polymer systems, using DOE to identify critical variables, improve yield/quality, and define reproducible process specifications for scale-up.
- Analyze analytical, thermal, and process data to recommend next-step experiments, operating ranges, and material-quality controls.

University of Colorado Boulder — July 2021 - July 2025

Graduate Researcher

- Designed and executed hypothesis-driven experiments across 15+ material variants, interpreting performance and degradation data to propose next-step design and process decisions.
- Authored clear technical conclusions from complex multi-technique datasets for multidisciplinary research and development stakeholders.

Georgia Institute of Technology — Aug 2019 - July 2021

Graduate Researcher

- Co-led thin-film process development for defense-oriented coatings, establishing reproducible deposition strategies and practical process-control parameters in a demanding R&D setting.

ETH Zurich — May 2017 - July 2017

Research Assistant

- Optimized polymer thin-film deposition processes with DOE and in-situ monitoring, improving process understanding and reproducibility.

University of Colorado Boulder — 2021 - 2025

Laboratory Operations Manager

- Managed operations for a 15-member research group, coordinated instrument planning and SOP-style documentation, and improved instrument uptime by 30%.

Georgia Institute of Technology — 2019 - 2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates and developed supplemental materials that improved exam performance by 15%, strengthening technical communication and training skills.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Bio-Based Polymer Scale-Up Process
Current

Leading DOE-guided process optimization and characterization studies that define reproducible process windows and quality-relevant specifications for polymer systems moving toward scale-up.

DOE, process development, scale-up, characterization

AFOSR Defense Thin-Film Coatings
DoD Funded

Developed and refined deposition processes for thin-film coatings, balancing experimental execution, data analysis, and practical process control decisions.

Thin films, process development, experimental design

Electrochromic Material Reliability Study
Chemistry of Materials 2025

Linked formulation-adjacent design choices with material performance using structured experimental plans and multi-technique characterization.

FT-IR, UV-Vis-NIR, cyclic voltammetry, TGA, DSC, data analysis

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Process Development: DOE, process optimization, protocol development, scale-up specifications, workplans, reproducibility studies, technical reporting
Analytical Characterization: FT-IR, UV-Vis-NIR, LC-MS, GC-MS, NMR, TGA, DSC, GPC/SEC, ellipsometry, QCM-D, cyclic voltammetry
Data & Quality Methods: SPC (Coursera 2026), JMP (basic), Python (basic), Excel (advanced), out-of-trend investigation, documentation, SOP-ready controls
Materials & Product Development: Polymers, thin films, structure-property-processing relationships, stability-sensitive materials, product-performance evaluation
Project Execution: Cross-functional communication, external-partner-ready technical summaries, multi-project prioritization, laboratory coordination

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)